Application of Information Theory, Lecture 2
Joint & Conditional Entropy, Mutual Information

Handout Mode

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Part I

Joint and Conditional Entropy

Joint entropy

- Recall that the entropy of rv X , is defined by

$$H(X) = -\sum_{x \in \mathcal{X}} \mathrm{P}_X(x) \log \mathrm{P}_X(x) = \mathop{\mathrm{E}}_{x \leftarrow X} H_X(x)$$

- Shorter notation: for $X \sim p$, let $H(X) = -\sum_x p(x) \log p(x)$ (where the summation is over the domain of X).
- The joint entropy of (jointly distributed) rvs X and Y with $(X, Y) \sim p$, is

$$H(X, Y) = -\sum_{x,y} p(x, y) \log p(x, y) = \mathop{\mathrm{E}}_{(x,y) \leftarrow (X,Y)} H_{X,Y}(x, y)$$

This is simply the entropy of the rv $Z = (X, Y)$.

- Example:

$X \backslash Y$	0	1
0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{2}$	0

$$\begin{aligned} H(X, Y) &= -\frac{1}{2} \log \frac{1}{\frac{1}{2}} - \frac{1}{4} \log \frac{1}{\frac{1}{4}} - \frac{1}{4} \log \frac{1}{\frac{1}{4}} \\ &= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 2 = 1\frac{1}{2} \end{aligned}$$

Joint entropy, cont.

- The joint entropy of $(X_1, \ldots, X_n) \sim p$, is

$$H(X_1, \ldots, X_n) = -\sum_{x_1, \ldots, x_n} p(x_1, \ldots, x_n) \log p(x_1, \ldots, x_n)$$

Conditional entropy

- Let $(X, Y) \sim p$, let $p_X = \sum_y p(x, y)$, $p_Y = \sum_x p(x, y)$ and $p_{Y|X}(y|x) = \frac{p(x,y)}{p_X(x)}$.
- Conditional sample entropy: $H_{Y|X}(y|x) = -\log p_{Y|X}(y|x)$
- For $x \in \mathrm{Supp}(X)$, the random variable $Y|_{X=x}$ is well defined (distributed according to $q(y) = p_{Y|X}(y|x)$).
- The entropy of Y conditioned on X, is defined by

$$H(Y|X) := \mathop{\mathrm{E}}_{x \leftarrow X} H(Y|_{X=x}) = \mathop{\mathrm{E}}_{(x,y) \leftarrow (X,Y)} H_{Y|X}(y|x)$$

- Measures the uncertainty in Y given X.
- $$\begin{aligned} H(Y|X) &= \mathop{\mathrm{E}}_{(x,y) \leftarrow (X,Y)} H_{Y|X}(y|x) = -\mathop{\mathrm{E}}_{(x,y) \leftarrow (X,Y)} \log p_{Y|X}(y|x) \\ &= -\mathop{\mathrm{E}}_{Z = p_{Y|X}(Y|X)} \log Z \end{aligned}$$

Conditional entropy, cont.

- Example

$X \backslash Y$	0	1
0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{2}$	0

What is $H(Y|X)$ and $H(X|Y)$?

$$\begin{aligned} H(Y|X) &= \mathop{\mathrm{E}}_{x\leftarrow X} H(Y|_{X=x}) \\ &= \frac{1}{2}H(Y|_{X=0}) + \frac{1}{2}H(Y|_{X=1}) \\ &= \frac{1}{2}H(\frac{1}{2},\frac{1}{2}) + \frac{1}{2}H(1,0) = \frac{1}{2}. \end{aligned}$$

$$\begin{aligned} H(X|Y) &= \mathop{\mathrm{E}}_{y\leftarrow Y} H(X|_{Y=y}) \\ &= \frac{3}{4}H(X|_{Y=0}) + \frac{1}{4}H(X|_{Y=1}) \\ &= \frac{3}{4}H(\frac{1}{3},\frac{2}{3}) + \frac{1}{4}H(1,0) = 0.6887 \neq H(Y|X). \end{aligned}$$

Conditional entropy, cont..

- $$\begin{aligned} H(X|Y,Z) &= \mathop{\mathrm{E}}_{(y,z)\leftarrow(Y,Z)} H(X|_{Y=y,Z=z}) \\ &= \mathop{\mathrm{E}}_{y\leftarrow Y} \mathop{\mathrm{E}}_{z\leftarrow Z|_{Y=y}} H(X|_{Y=y,Z=z}) \\ &= \mathop{\mathrm{E}}_{y\leftarrow Y} \mathop{\mathrm{E}}_{z\leftarrow Z|_{Y=y}} H((X|_{Y=y})|_{Z=z}) \end{aligned}$$

Let $(X_y, Z_y) = (X,Z)|_{Y=y}$. Then

$$\begin{aligned} H(X|Y,Z) &= \mathop{\mathrm{E}}_{y\leftarrow Y} \mathop{\mathrm{E}}_{z\leftarrow Z_y} H(X_y|_{Z=z}) \\ &= \mathop{\mathrm{E}}_{y\leftarrow Y} \mathop{\mathrm{E}}_{z\leftarrow Z_y} H(X_y|_{Z_y=z}) \\ &= \mathop{\mathrm{E}}_{y\leftarrow Y} H(X_y|Z_y) \end{aligned}$$

Relating mutual entropy to conditional entropy

- What is the relation between $H(X)$, $H(Y)$, $H(X,Y)$ and $H(Y|X)$?
- Intuitively, $0 \leq H(Y|X) \leq H(Y)$

  Non-negativity is immediate. We prove upperbound later.
- We will also see that $H(Y|X) = H(Y)$ iff X and Y are independent.
- In our example, $H(Y) = H(\frac{3}{4}, \frac{1}{4}) > \frac{1}{2} = H(Y|X)$
- Note that $H(Y|_{X=x})$ might be larger than $H(Y)$ for some $x \in \mathrm{Supp}(X)$.
- Chain rule (proved next). $H(X,Y) = H(X) + H(Y|X)$
- Intuitively, uncertainty in (X,Y) is the uncertainty in X plus the uncertainty in Y given X.
- $H(Y|X) = H(X,Y) - H(X)$ is as an alternative definition for $H(Y|X)$.

Chain rule (for the entropy function)

Claim 1

For rvs X, Y, it holds that $H(X,Y) = H(X) + H(Y|X)$.

- Proof immediately follow by the grouping axiom:

$X \backslash Y$			
	$P_{1,1}$	$\cdots$	$P_{1,n}$
	$\vdots$	$\vdots$	$\vdots$
	$P_{n,1}$	$\cdots$	$P_{n,n}$

Let $q_i = \sum_{j=1}^{n} p_{i,j}$ $\quad (= \Pr[X = i])$

$$
\begin{aligned}
&H(P_{1,1}, \ldots, P_{n,n}) \\
&= H(q_1, \ldots, q_n) + \sum_i q_i H\left(\frac{P_{i,1}}{q_i}, \ldots, \frac{P_{i,n}}{q_i}\right) \\
&= H(X) + H(Y|X).
\end{aligned}
$$

- Another proof. Let $(X,Y) \sim p$, and recall that $p(x,y) = p_X(x) \cdot p_{Y|X}(y|x)$.

$\implies \log p(x,y) = \log p_X(x) + \log p_{Y|X}(y|x)$

$\implies \mathrm{E} \log p(X,Y) = \mathrm{E} \log p_X(X) + \mathrm{E} \log p_{Y|X}(Y|X)$

$\implies H(X,Y) = H(X) + H(Y|X)$.

$H(Y|X) \leq H(Y)$

Jensen inequality: for any concave function f, values $t_1, \ldots, t_k$ and $\lambda_1, \ldots, \lambda_k \in [0,1]$ with $\sum_i \lambda_i = 1$, it holds that $\sum_i \lambda_i f(t_i) \leq f(\sum_i \lambda_i t_i)$.
Let $(X, Y) \sim p$.

$$
\begin{aligned}
H(Y|X) &= -\sum_{x,y} p(x,y) \log p_{Y|X}(y|x) \\
&= \sum_{x,y} p(x,y) \log \frac{p_X(x)}{p(x,y)} \\
&= \sum_{x,y} p_Y(y) \cdot \frac{p(x,y)}{p_Y(y)} \log \frac{p_X(x)}{p(x,y)} \\
&= \sum_{y} p_Y(y) \sum_{x} \frac{p(x,y)}{p_Y(y)} \log \frac{p_X(x)}{p(x,y)} \\
&\leq \sum_{y} p_Y(y) \log \sum_{x} \frac{p(x,y)}{p_Y(y)} \frac{p_X(x)}{p(x,y)} \\
&= \sum_{y} p_Y(y) \log \frac{1}{p_Y(y)} = H(Y).
\end{aligned}
$$

$H(Y|X) \leq H(Y)$ cont.

- Assume X and Y are independent (i.e., $p(x,y) = p_X(x) \cdot p_Y(y)$ for any x, y)

$\Longrightarrow$ $p_{Y|X}(y|x) = p_Y(y)$ for any x, y

$\Longrightarrow$ $H(Y|X) = H(Y)$

- Is the converse also true: $H(Y|X) = H(Y)$ implies X and Y are independent?

  Yes, since log is strictly concave in the range. Equality happens iff all t_i (In Jensen inequality) are the same,

- which happens iff $p(x,y) = p_X(x)p_Y(y)$ for all x, y

Other inequalities

- $H(X), H(Y) \leq H(X,Y) \leq H(X) + H(Y)$.

  Follows from $H(X,Y) = H(X) + H(Y|X)$.
  - Left inequality since $H(Y|X)$ is non negative.
  - Right inequality since $H(Y|X) \leq H(Y)$.
- $H(X,Y|Z) = H(X|Z) + H(Y|X,Z)$ (by chain rule)
- $H(X|Y,Z) \leq H(X|Y)$

Proof:

$$\begin{aligned}
H(X|Y,Z) &= \mathop{\mathrm{E}}_{(z,y)\leftarrow(Z,Y)} H(X|_{(Y,Z)=(y,z)}) \\
&= \mathop{\mathrm{E}}_{y\leftarrow Y} \mathop{\mathrm{E}}_{z\leftarrow Z|_{Y=y}} H(X|_{(Y,Z)=(z,y)}) \\
&= \mathop{\mathrm{E}}_{y\leftarrow Y} \mathop{\mathrm{E}}_{z\leftarrow Z|_{Y=y}} H((X|_{Y=y})|_{Z=z}) \\
&= \mathop{\mathrm{E}}_{y\leftarrow Y} H(X|_{Y=y}|Z|_{Y=y}) \\
&\leq \mathop{\mathrm{E}}_{y\leftarrow Y} H(X|_{Y=y}) \\
&= H(X|Y).
\end{aligned}$$

Chain rule (for the entropy function), general case

Claim 2

For rvs $X_1, \ldots, X_k$, it holds that
$H(X_1, \ldots, X_k) = H(X_1) + H(X_2|X_1) + \ldots + H(X_k|X_1, \ldots, X_{k-1})$.

Proof: ?

- Extremely useful property!
- Analogously to the two variables case, it also holds that:
- $H(X_i) \leq H(X_1, \ldots, X_k) \leq \sum_i H(X_i)$
- $H(X_1, \ldots, X_K|Y) \leq \sum_i H(X_i|Y)$

Examples

- (from last class) Let $X_1, \ldots, X_n$ be Boolean iid with $X_i \sim (\frac{1}{3}, \frac{2}{3})$. Compute $H(X_1, \ldots, X_n)$
- As above, but X_n is set to $\bigoplus_{1 \leq i \leq n-1} X_i$?
  - Via chain rule?
  - Via mapping?

Applications

- Let $X_1, \ldots, X_n$ be Boolean iids with $X_i \sim (p, 1-p)$ and let $X = X_1, \ldots, X_n$. Let f be such that $\Pr[f(X) = z] = \Pr[f(X) = z']$, for every $k \in \mathbb{N}$ and $z, z' \in \{0,1\}^k$. Let $K = |f(X)|$.

  Prove that $\mathrm{E}\, K \leq n \cdot h(p)$.

- $$\begin{aligned} n \cdot h(p) &= H(X_1, \ldots, X_n) \\ &\geq H(f(X), K) \\ &= H(K) + H(f(X) \mid K) \\ &= H(K) + \mathrm{E}\, K \\ &\geq \mathrm{E}\, K \end{aligned}$$

- Interpretation
- Upper bounds

Applications cont.

- How many comparisons it takes to sort n elements?

  Let S be a sorter for n elements making t comparisons.

  What can we say about t?

- Let X be a uniform random permutation of $[n]$ and let $Y_1, \ldots, Y_t$ be the answers S gets when sorting X.
- X is determined by $Y_1, \ldots, Y_t$.

  Namely, $X = f(Y_1, \ldots, Y_t)$ for some function f.
- $H(X) = \log n!$
- $$\begin{aligned} H(X) &= H(f(Y_1, \ldots, Y_t)) \\ &\leq H(Y_1, \ldots, Y_t) \\ &\leq \sum_i H(Y_i) \\ &\leq t \end{aligned}$$

$\Longrightarrow \quad t \geq \log n! = \Theta(n \log n)$

Concavity of entropy function

Let $p = (p_1, \ldots, p_n)$ and $q = (q_1, \ldots, q_n)$ be two distributions, and for $\lambda \in [0, 1]$ consider the distribution $\tau_\lambda = \lambda p + (1 - \lambda)q$.
(i.e., $\tau_\lambda = (\lambda p_1 + (1 - \lambda)q_1, \ldots, \lambda p_n + (1 - \lambda)q_n)$).

Claim 3

$H(\tau_\lambda) \geq \lambda H(p) + (1 - \lambda)H(q)$

Proof:

- Let Y over $\{0, 1\}$ be 0 wp λ
- Let X be distributed according to p if $Y = 0$ and according to q otherwise.
- $H(\tau_\lambda) = H(X) \geq H(X \mid Y) = \lambda H(p) + (1 - \lambda)H(q)$

We are now certain that we drew the graph of the (two-dimensional) entropy function right...

Part II

Mutual Information

Mutual information

- $I(X;Y)$ — the "information" that X gives on Y
- $$\begin{aligned} I(X;Y) &:= H(Y) - H(Y|X) \\ &= H(Y) - (H(X,Y) - H(X)) \\ &= H(X) + H(Y) - H(X,Y) \\ &= H(X) - H(X|Y) \\ &= I(Y;X). \end{aligned}$$
- The mutual information that X gives about Y equals the mutual information that Y gives about X.
- $I(X;Y) \geq 0$. When 0?
- $I(X;X) = H(X)$
- $I(X;f(X)) = H(f(X))$ (and smaller than $H(X)$ if f is non-injective)
- $I(X;Y,Z) \geq I(X;Y), I(X;Z)$ (since $H(X \mid Y,Z) \leq H(X \mid Y), H(X \mid Z)$)
- $I(X;Y|Z) := H(Y|Z) - H(Y|X,Z) \geq 0$
- $I(X;Y|Z) = I(Y;X|Z)$ (since $I(X';Y') = I(Y';X')$)

Numerical example

- Example

$X \backslash Y$	0	1
0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{2}$	0

$$
\begin{aligned}
I(X;Y) &= H(X) - H(X|Y) \\
&= 1 - \frac{3}{4} \cdot h(\frac{1}{3}) \\
&= I(Y;X) \\
&= H(Y) - H(Y|X) \\
&= h(\frac{1}{4}) - \frac{1}{2} h(\frac{1}{2})
\end{aligned}
$$

Chain rule for mutual information

Claim 4 (Chain rule for mutual information)

For rvs $X_1, \ldots, X_k, Y$, it holds that
$I(X_1, \ldots, X_k; Y) = I(X_1; Y) + I(X_2; Y|X_1) + \ldots + I(X_k; Y|X_1, \ldots, X_{k-1})$.

Proof: ? HW

Examples

- Let $X_1, \ldots, X_{n-1}$ be iid uniform bits (i.e., $X_i \sim (\frac{1}{2}, \frac{1}{2})$), and let $X_n = \bigoplus_{i \in [n-1]} X_i$. Compute $I(X_1, \ldots, X_{n-1}; X_n)$.
  - Directly, $I(X_1, \ldots, X_{n-1}; X_n) = H(X_n) - H(X_n | X_1, \ldots, X_{n-1}) = 1 - 0 = 1$
  - Using chain rule,

$$\begin{aligned} & I(X_1, \ldots, X_{n-1}; X_n) \\ & = I(X_1; X_n) + I(X_2; X_n | X_1) + \ldots + I(X_{n-1}; X_n | X_1, \ldots, X_{n-2}) \\ & = 0 + 0 + \ldots + 1 = 1. \end{aligned}$$

- Let T and F be the top and front side, respectively, of a 6-sided fair dice. Compute $I(T; F)$.

$$\begin{aligned} I(T; F) &= H(T) - H(T|F) \\ &= \log 6 - \log 4 \\ &= \log 3 - 1. \end{aligned}$$

Part III

Data Processing

Data processing inequality

Definition 5 (Markov Chain)

$(X, Y, Z) \sim p$ form a Markov chain, denoted $X \to Y \to Z$, if $p_{Z|X,Y}(z|x, y) = p_{Z|Y}(z|y)$, for all $(x, y, z) \in \text{Supp}(X, Y, Z)$.

Example: random walk on graph.

Claim 6

If $X \to Y \to Z$, then $I(X; Y) \geq I(X; Z)$.

- $$H(Z|X, Y) = \mathop{\mathrm{E}}_{y \leftarrow Y}\left[\mathop{\mathrm{E}}_{x \leftarrow X|_{Y=y}}[H((Z|_{Y=y})|_{X=x})]\right]$$
$$= \mathop{\mathrm{E}}_{y \leftarrow Y}[H(Z|_{Y=y})] = H(Z|Y)$$
- Hence, $I(X; Z|Y) = H(Z|Y) - H(Z|Y, X) = 0$.
- By Chain rule, $I(X; Y, Z) = I(X; Z) + I(X; Y|Z) = I(X; Y) + I(X; Z|Y)$.
- Since $I(X; Y|Z) \geq 0$, we conclude $I(X; Y) \geq I(X; Z)$.□

Fano's Inequality

- How well can we guess X from Y?
- Could with no error if $H(X|Y) = 0$. What if $H(X|Y)$ is small?

Theorem 7 (Fano's inequality)

For any rvs X and Y, and any (even random) g, it holds that

$$h(P_e) + P_e \log |\mathcal{X}| \geq H(X|\hat{X}) \geq H(X|Y)$$

for $\hat{X} = g(Y)$ and $P_e = \Pr\left[\hat{X} \neq X\right]$.

- Note that $P_e = 0$ implies that $H(X|Y) = 0$
- The inequality can be weakened to $1 + P_e \log |\mathcal{X}| \geq H(X|Y)$,
- Alternatively, to $P_e \geq \frac{H(X|Y)-1}{\log|\mathcal{X}|}$
- Intuition for $\propto \frac{1}{\log|\mathcal{X}|}$
- We call $\hat{X}$ an estimator for X (from Y).

Proving Fano's inequality

Let X and Y be rvs, let $\hat{X} = g(Y)$ and $P_e = \Pr\left[\hat{X} \neq X\right]$.

- Let $D = \begin{cases} 1, & \hat{X} \neq X \\ 0, & \hat{X} = X. \end{cases}$

$$H(D, X|\hat{X}) = H(X|\hat{X}) + \underbrace{H(D|X, \hat{X})}_{=0}$$

$$= \underbrace{H(D|\hat{X})}_{\leq H(D)=h(P_e)} + \underbrace{H(X|D, \hat{X})}_{\leq P_e \log|\mathcal{X}|(?)}$$

- It follows that $h(P_e) + P_e \log|\mathcal{X}| \geq H(X|\hat{X})$
- Since $X \to Y \to \hat{X}$, it holds that $I(X; Y) \geq I(X; \hat{X})$

  $\implies H(X|\hat{X}) \geq H(X|Y)$